Table 2 Overview of included studies on HRQOL measures

Region	Study Author year	Study design/ study type	Patient population	Treatment	Follow-up times assessed	Assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
HRQOL measures mappable to direct utility values									
FACT-G									
Global	JULIET Maziarz 2020 [19] Tam 2019 [67] Maziarz 2017 [68]	Phase 2 single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Tisagenlecleucel	Baseline 3 months 6 months 12 months 18 months	FACT-G TS	0 to 108 (higher scores reflecting better health)	77.4 (16.1)	+ 10 (11.1)
Global	ORCHARRD ClinicalTrials.gov [39]	Phase 3 trial	Adult patients with R/R DLBCL	Ofatumumab + DHAP R-DHAP	Baseline End of treatment	FACT-G TS	0 to 108 (higher scores reflecting better health)	NR NR	− 2.561 (SE, 0.7671) − 2.591 (SE, 0.7696)
Global	SADAL Shah 2021 [38] Casasnovas 2020 [69]	Phase 2b single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Selinexor	Baseline Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6 Cycle 7 End of treatment	FACT-G	0 to 116 (higher scores reflecting better health)	Responders: 72.6 (95% CI, 70.9, 74.3) Nonresponders: 72.9 (95% CI, 71.3, 74.4)	Responders: − 6.0 (95% CI, − 9.2, − 2.8) Nonresponders: − 9.2 (NR)
US	NCT00667615 Straus 2015 [40]	Phase 1/2 single-arm trial	Patients with R/R DLBCL, aged 60 years or older, not eligible for autologous HSCT	R-CVEP	Baseline Cycle 1 Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6 3 months 6 months 9 months 12 months	FACT-G TS	0 to 108 (higher scores reflecting better health)	75.48 (17.44)	+ 11.72 (15.14)
FACT-Lym									

Table 2 (continued)

Region	Study Author year	Study design/ study type	Patient population	Treatment	Follow-up times assessed	Assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
Global	JULIET Maziarz 2020 [19] Tam 2019 [67] Maziarz 2017 [68]	Phase 2 single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Tisagenlecleucel	Baseline 3 months 6 months 12 months 18 months	FACT-Lym TS[a]	0 to 168 (higher scores reflecting better health)	121.2 (24.0)	+ 13.1 (16.1)
						FACT-Lym TOI	0 to 116 (higher scores reflecting better health)	82.0 (19.0)	+ 9.2 (13.6)
						FACT-Lym subscale	0 to 60 (higher scores reflecting better health)	44.4 (9.1)	+ 3.1 (6.6)
Global	ORCHARRD ClinicalTrials.gov [39]	Phase 3 trial	Adult patients with R/R DLBCL	Ofatumumab + DHAP R-DHAP	Baseline End of treatment	FACT-Lym TOI	0 to 116 (higher scores reflecting better health)	NR	− 2.028 (SE, 0.9196) − 3.156 (SE, 0.9204)
Global	SADAL Shah 2021 [38] Casasnovas 2020 [69]	Phase 2b single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Selinexor	Baseline Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6 Cycle 7 End of treatment	FACT-Lym TS	0 to 168 (higher scores reflecting better health)	Responders: 116.7 (95% CI, 114.2, 119.2) Nonresponders: 117.1 (95% CI, 114.9, 119.4)	Responders: − 9.9 (95% CI, − 15.5, − 4.3) Nonresponders: − 15.7 (NR)
						FACT-Lym TOI	0 to 116 (higher scores reflecting better health)	Responders: 77.4 (95% CI, 75.3, 79.4) Nonresponders: 79.2 (95% CI, 77.3, 81.0)	Responders: − 8.6 (95% CI, − 13.0, − 4.3) Nonresponders: − 13.5 (NR)

Table 2 (continued)

Region	Study Author year	Study design/ study type	Patient population	Treatment	Follow-up times assessed	Assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
US	NCT00667615 Straus 2015 [40]	Phase 1/2 single-arm trial	Patients with R/R DLBCL, aged 60 years or older, not eligible for autologous HSCT	R-CVEP	Baseline Cycle 1 Cycle 2 Cycle 3 Cycle 4 Cycle 5 Cycle 6 3 months 6 months 9 months 12 months	FACT-Lym TS[b]	0 to 168 (higher scores reflecting better health)	117.1 (26.29)	+ 18.61 (19.89)
						FACT-Lym TOI	0 to 116 (higher scores reflecting better health)	76 (21.48)	+ 12.29 (19.49)
NR	TRANSFORM Abramson 2021 [18]	Phase 3 RCT	Adults aged ≤ 75 years with R/R LBCL ≤ 12 months after 1L therapy	Liso-cel	Baseline 1 month 2 months 3 months 6 months End of study or 36 months	FACT-Lym subscale	0 to 60 (higher scores reflecting better health)	NR	+ 1.48 (95% CI, − 0.30, 3.26)[c]
				SOC (salvage chemotherapy + BEAM + autologous HSCT)				NR	+ 1.63 (95% CI, − 0.41, 3.68)
EORTC QLQ-C30									
UK, US, France, Germany, Italy, Spain	Ma 2021 [41]	Point in time survey	Adult patients with DLBCL in 2 or 3 + lines of therapy	NR	NR	EORTC QLQ-C30 global health status[d]	0 to 100 (higher scores reflecting better health)	2L: 55.48 (18.02) 3L +: 49.93 (21.07)	NR
US	TRANSCEND NHL 001 Patrick 2021 [17] Patrick 2020 [37] Patrick 2019 [35] Patrick 2019 [36]	Phase 1 single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Liso-cel	Baseline 1 month 2 months 3 months 6 months 9 months 12 months 18 months	EORTC QLQ-C30 global health status[c]	0 to 100 (higher scores reflecting better health)	62.3 (20.3)	19.7 (25.6)
NR	TRANSFORM Abramson 2021 [18]	Phase 3 single-arm trial	Adults aged ≤ 75 years with R/R LBCL ≤ 12 months after 1L therapy	Liso-cel	Baseline 1 month 2 months 3 months 6 months End of study or 36 months	EORTC QLQ-C30 global health status[d]	0 to 100 (higher scores reflecting better health)	NR	+ 3.08 (95% CI, − 1.83, 7.99)
				SOC (salvage chemotherapy + BEAM + autologous HSCT)				NR	+ 0.04 (95% CI, − 5.24, 5.31)[c]

Table 2 (continued)

Region	Study Author year	Study design/ study type	Patient population	Treatment	Follow-up times assessed	Assessment method	Scale	Baseline mean (SD)	Mean change at last follow-up (SD)
SF-36									
Global	JULIET Maziarz 2020 [19] Tam 2019 [67] Maziarz 2017 [68]	Phase 2 single-arm trial	Adult patients with R/R DLBCL with 2 + prior lines of therapy	Tisagenlecleucel	Baseline 3 months 6 months 12 months 18 months	SF-36 PCS	0 to 100 (higher scores reflecting better health)	44.3 (9.17)	+ 4.3 (10.6)
						SF-36 MCS	0 to 100 (higher scores reflecting better health)	48.1 (10.5)	+ 2.3 (9.8)

1L first line; *2L* second line; *3L+* third line or later; *BEAM* carmustine, etoposide, cytarabine, and melphalan; *CI* confidence interval; *DHAP* dexamethasone, cytarabine, and cisplatin; *DLBCL* diffuse large B cell lymphoma; *EORTC QLQ-C30* European Organisation for Research and Treatment of Cancer Quality of Life Questionnaire C30; *FACT-G* Functional Assessment of Cancer Therapy—General; *FACT-Lym* Functional Assessment of Cancer Therapy—Lymphoma; *FACT-Lym TOI* Functional Assessment of Cancer Therapy—Lymphoma Trial Outcomes Index; *HRQOL* health-related quality of life; *HSCT* hematopoietic stem cell transplantation; *LBCL* large B cell lymphoma; *liso-cel* lisocabtagene maraleucel; *MCS* mental component summary; *NR* not reported; *PCS* physical component summary; *RCT* randomized controlled trial; *R-CVEP* rituximab plus cyclophosphamide, etoposide, and prednisone; *R-DHAP* rituximab plus dexamethasone, cytarabine, and cisplatin; *R/R* relapsed or refractory; *SD* standard deviation; *SE* standard error; *SF-36* 36-item Short Form; *SOC* standard of care; *TS* total score; *UK* United Kingdom; *US* United States

[a]FACT-Lym TS is composed of the FACT-G score (0−108) and FACT-Lym subscale (0−60)

[b]Utility values can occasionally be below zero, reflecting a health state less preferable than dead

[c]Study also reported select components of the EORTC QLQ-C30 symptomatic and functional domains

[d]Reported value is overall least squares mean changes from baseline to 2 months after liso-cel therapy